Adaptive Cognitive Load Prediction System for Online Learning Using Behavioral Patterns

COMMON INTEGRATED SOLUTION REPORT

Team member	Student ID	Individual component
Imesh L A R	IT22281364	Learner profile and cognitive style detection
Subasinghe M.H.R.I	IT22089786	Real time cognitive load prediction
Jayasinghe J. M. S. N	IT22166074	Resource upload and adaptive content conversion
Dilshan W.G.W	IT22055712	Explainable support recommendations and assessment

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science Honours in Information Technology

Specialization in Software Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

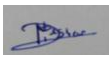
October 2026

Group ID R26-SE-004

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. Generative AI assistance was used to organise and edit this common report and check bibliographic details. We remain responsible for its accuracy, originality, citations, technical descriptions, analyses, and conclusions. The team will confirm the final disclosure in accordance with the applicable module policy before submission.

Student name	Student ID	Signature	Date
Imesh L A R	IT22281364		11/10/2026
Subasinghe M.H.R.I	IT22089786		11/10/2026
Jayasinghe J. M. S. N	IT22166074		11/10/2026
Dilshan W.G.W	IT22055712		11/10/2026

Supervisor certification. The above candidates carried out this research project under my supervision.

Role and name	Signature	Date
Supervisor: Dr Kalpani Manathunga		
Co-supervisor: Mr Jeewaka Perera		

ABSTRACT

Online learning environments can deliver instructional resources and record student activity while providing limited support for interpreting learning effort or adapting material to individual needs. Lumora addresses this problem through an integrated platform that connects learner profiling and cognitive style, real-time cognitive-load prediction, resource preparation and adaptive content conversion, and explainable teacher and learner support. The solution combines questionnaire-based learner information, webcam and cursor observations, six video-interaction features, source-grounded generation, LIME and SHAP explanations, and teacher review. A cognitive-load study involved 171 undergraduate computing students and produced 1,420 valid two-minute observation windows. A participant-separated evaluation of 293 windows achieved 79.52% accuracy and 79.55% macro F1 with Random Forest. Resource-processing verification passed 18 of 19 focused checks, with a mathematical PDF fixture exposing incomplete symbol preservation. Separate automated suites verified prediction, guidance, recommendation, assessment, and cognitive-style explanation functions. In preliminary blinded comparisons, three ICT teachers preferred system guidance in nine of nine scenario comparisons, with mean ratings of 4.84/5 against 3.87/5 for baseline guidance. A classroom pilot involving ten Grade 8 learners recorded 78 correct responses from 100 quiz attempts and a mean MCQ clarity rating of 4.30/5. These findings support technical feasibility and preliminary perceived usefulness, while the different populations, limited evaluations, model-version differences, and absence of a controlled learning-outcome study constrain generalization. The principal contribution is a traceable workflow linking learner evidence and educator resources to reviewable educational support. Further work should align feature definitions, verify extracted mathematics, strengthen model provenance, and evaluate learning outcomes across broader settings.

Keywords: adaptive learning; cognitive load; learner profiling; explainable artificial intelligence; source grounded content

ACKNOWLEDGEMENTS

We express our appreciation to Dr Kalpani Manathunga and Mr Jeewaka Perera for their supervision and guidance throughout the research project. We also acknowledge the Sri Lanka Institute of Information Technology and the Department of Computer Science and Software Engineering for the academic environment supporting this work.

We thank the undergraduate participants who contributed behavioral observations and mental-effort ratings, and the ICT teachers and classroom participants who contributed to the preliminary evaluations. Their involvement made it possible to investigate both the technical operation and the practical interpretation of the learning platform. We also appreciate the cooperation of the research team in bringing the four component workflows into a common solution.

TABLE OF CONTENTS

1 INTRODUCTION AND PROJECT CONTEXT 1
1.1 Background and context 1
1.2 Stakeholders and user needs 1
1.3 Alignment with the Sustainable Development Goals 2
1.4 Purpose and scope of the common report 2
2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS 4
2.1 Overall problem definition 4
2.2 Root causes and overall research gap 4
2.3 Subproblem and component mapping 5
2.4 Overall aim and group objectives 6
2.5 Scope assumptions and constraints 6
3 OVERALL SOLUTION STRUCTURE 8
3.1 Integrated solution overview 8
3.2 High level architecture 8
3.3 Component 1 - Learner profile and cognitive style detection 9
3.4 Component 2 - Real time cognitive load prediction 10
3.5 Component 3 - Resource upload and adaptive content conversion 11
3.6 Component 4 - Explainable support recommendations and assessment 11
3.7 End to end user and data flow 12
4 COMPONENT INTEGRATION AND COHESIVE OPERATION 14
4.1 Integration strategy 14
4.2 Interfaces data contracts and shared models 14
4.3 Shared nonfunctional requirements 16
4.4 Integrated workflow and exception handling 16
4.5 Team contribution and dependency matrix 18
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION 20
5.1 Common technology stack and infrastructure 20
5.2 Development and integration process 21
5.3 Deployment and operational configuration 22
5.4 Evidence of team execution 22
6 INTEGRATED TESTING EVALUATION AND VALIDATION 24
6.1 Evaluation framework and success criteria 24
6.2 Integration and end to end testing 25

6.3 Overall results 26
6.4 Stakeholder and user validation 30
6.5 Achievement of common objectives 33
6.6 Integrated limitations and threats 34
7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION 37
7.1 Value proposition and beneficiaries 37
7.2 Market alternatives and differentiation 37
7.3 Business model and feasibility 38
7.4 Adoption intellectual property and compliance 38
7.5 Sustainable Development Goal contribution 39
8 CONCLUSION AND FUTURE DIRECTIONS 40
8.1 Overall conclusion 40
8.2 Principal integrated contribution 40
8.3 Future directions 41
REFERENCES 43
APPENDICES 45
Appendix A Integrated interface specification 45
Appendix B Team contribution declaration 47
Appendix C Extended evaluation records 47

LIST OF FIGURES

Figure 1 Integrated platform architecture and service boundaries 9
Figure 2 Four component system flow and feedback 13
Figure 3 Resource preparation states and recovery 18
Figure 4 Cognitive load classifier accuracy and macro F1 29
Figure 5 Teacher guidance criterion means across nine paired cases 31
Figure 6 Adaptive MCQ ratings and answer key review boundary 32

LIST OF TABLES

Table 1 Four subproblems and component handoffs 5
Table 2 Principal shared records 15
Table 3 Failure handling across the integrated workflow 17
Table 4 Team responsibilities and dependencies 18
Table 5 Shared technology stack 20
Table 6 Evaluation contexts and evidence boundaries 24
Table 7 Recorded automated verification by group 25
Table 8 Cognitive Style classifier comparison 27
Table 9 Class-wise cognitive style classification performance 27
Table 10 Cognitive load classifier comparison in percentage units 28
Table 11 Cognitive load class recall on held out windows 29
Table 12 Teacher guidance comparison across nine paired cases 30
Table 13 Descriptive expert ratings of MCQ sets 31
Table 14 Exploratory classroom pilot summary 32
Table 15 Common objective achievement and remaining validation 34
Table 16 Integrated risks and priority corrective actions 35
Table 17 Gateway interface responsibilities 45
Table 18 Six feature inference contract and reliability field 46
Table 19 Anonymized classroom learner observations 47
Table 20 Evaluation denominators and interpretation 48

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
ASR	Automatic speech recognition
ASSIST	Approaches and Study Skills Inventory for Students
CLT	Cognitive Load Theory
CoVe	Chain of Verification
CSV	Comma separated values
FIFO	First in first out
FRE	Flesch Reading Ease
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
LIME	Local Interpretable Model agnostic Explanations
LLM	Large language model
MCQ	Multiple choice question
ML	Machine learning
OMML	Office Math Markup Language
PDF	Portable Document Format
PPTX	Office Open XML presentation format
RF	Random Forest
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over sampling Technique
SVM	Support Vector Machine
UTC	Coordinated Universal Time
XAI	Explainable artificial intelligence

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Digital learning platforms make videos, presentations, documents, and quizzes available through a common interface. However, access to material does not establish that a learner can process it effectively. The same lesson can impose different demands depending on prior knowledge, task complexity, interruptions, and presentation. Cognitive Load Theory explains why instructional design should consider limited processing capacity and the development of knowledge structures [1]. This motivates a system that considers learning effort alongside ordinary course progress.

Lumora is an adaptive learning platform developed under research group R26-SE-004. Its purpose is to connect student interaction evidence with educator-controlled content and understandable support. Four components operate in a defined flow: learner profile and cognitive-style detection; real-time cognitive-load prediction; resource upload and adaptive content conversion; and explanations, recommendations, and adaptive assessment. The platform retains a distinction between a profile describing selected learner characteristics and a changing load estimate associated with a learning session.

The solution uses behavioral observations as contextual evidence. It does not measure mental effort directly or assign an immutable learner identity. Generated outputs are educational support rather than autonomous instructional authority. The integrated design therefore includes traceable records, explicit insufficient data states, technical evidence views, and teacher decisions around guidance and lesson recommendations.

1.2 Stakeholders and user needs

Students need access to prepared lessons, clear progress information, manageable explanations, and feedback on their current learning activity. Their interaction with the platform should remain understandable even when a prediction is skipped or an external service is unavailable. Presentation choices should complement, rather than restrict, access to the original material. Learners also need to understand why webcam permission is requested during profiling and how their behavioral records are used.

Teachers need a practical way to organize courses, prepare heterogeneous materials, identify possible difficulty, and review proposed support. A raw category such as High does not explain which observed behaviors influenced the model. Teachers therefore require plain-language explanations with optional technical evidence, source-linked content, review controls, and a record of changes. Administrators need to approve educator accounts and oversee authorized access. Technical maintainers need consistent identifiers, model metadata, operational status, and controlled recovery from failed processing.

Stakeholder evidence has different scopes. Undergraduate data collection supports behavioral studies, while the three-teacher comparisons address perceived guidance and MCQ suitability. The ten-learner classroom pilot contributes a limited example of hosted use.

1.3 Alignment with the Sustainable Development Goals

The project is most closely aligned with Sustainable Development Goal 4, which concerns inclusive and equitable quality education and lifelong learning [2]. Browser-based delivery and the use of ordinary interaction traces make the design relevant to access to learning support without specialized sensing equipment. Reviewable explanations and lesson-grounded assessments also support the intention to improve the quality of digital educational experiences.

This alignment is an intended contribution. The evaluation does not demonstrate changes in educational access, attainment, or inequality. Realizing that contribution requires accessible navigation, language support, reasonable bandwidth demands, device compatibility, and teacher training. Webcam-based profiling may exclude some learners unless a suitable alternative is offered. Sustainable deployment should therefore assess both the benefits of adaptive support and the barriers created by its technical dependencies.

1.4 Purpose and scope of the common report

This report presents the complete solution as one research and software engineering contribution. It defines the shared problem, explains the four component boundaries in system-flow order, documents their interfaces, and synthesizes implementation and

evaluation findings. The focus is how learner evidence, instructional resources, prediction, explanation, and human review work together.

The report separates implemented mechanisms from empirical findings and proposed improvements. Model accuracy, function-level verification, teacher preference, and classroom quiz performance answer different questions and are not combined into one success percentage. Chapters 2 and 3 define the problem and architecture; Chapters 4 and 5 explain integration and execution; Chapter 6 evaluates the evidence; Chapter 7 considers sustainability and adoption; and Chapter 8 presents conclusions and future work. Appendices document interfaces, team responsibilities, and extended evaluation records.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem is that online learning activity and instructional resources are not consistently connected to timely, interpretable, and learner-responsive support. A platform may retain videos and activity logs without distinguishing insufficient activity from meaningful evidence of effort. It may hold profile information without using it carefully in presentation. It may generate fluent content without preserving the educator's intended facts or mathematical expressions. It may also display analytical labels without explaining their basis or giving teachers a reviewable response.

Addressing this problem requires a connected workflow rather than an isolated classifier or chatbot. The shared research question is how behavioral learner information and educator resources can be transformed into traceable adaptive support while keeping uncertainty, source integrity, and teacher responsibility visible. Successful integration means that each downstream operation receives the correct learner context and source evidence, not merely that every service can return an output.

2.2 Root causes and overall research gap

The first root cause is fragmented learner information. Approaches to studying, presentation-related observations, and temporary learning effort are different constructs. Research on approaches to studying provides a basis for distinguishing Deep, Strategic, and Surface approaches [3], while analytic and holistic thinking requires its own interpretation [4]. Combining labels without preserving these distinctions can make a profile appear more certain than its measurements justify.

The second cause is the ambiguity of behavioral traces. A pause or reply can reflect difficulty, deliberate review, distraction, or interest. Research on online video pausing reinforces the need to interpret activity within context [5]. The third cause is resource fragmentation: overlapping slides, documents, and speech can introduce redundancy, while extraction can damage symbols and omit qualifications. Research on generative summarization demonstrates that fluent output may contain information unsupported by its input [6].

The final cause is the gap between machine output and educational action. Local explanation methods describe a model's behavior but do not establish the psychological cause of a learner's effort. Recommendations and assessments add further interpretive steps. The research gap addressed by Lumora is the integration of these stages with explicit contracts, stage-specific validation, and review. It is not a claim that adaptive learning, behavioral classification, or explainable AI has no prior research.

Matching instruction to a style label also requires caution. Evidence reviews do not justify if every learner benefits from material matched to a presumed learning style [7]. In this project, profile information conditions presentation; its educational benefit remains an empirical question. This distinction prevents classification agreement from being treated as proof of instructional effectiveness.

2.3 Subproblem and component mapping

Four subproblems define the division of work. Component 1 develops a structured learner profile from questionnaire responses and behavior associated with visual and verbal content. Component 2 estimates changing cognitive load from video interactions. Component 3 prepares educator resources and adapts their presentation using learner-state inputs. Component 4 translates predictions into explanations, guidance, lesson planning, and assessment. Table 1 maps these responsibilities and their downstream dependencies.

Table 1 Four subproblems and component handoffs

Component	Subproblem	Output and dependency
1 Profiling	Represent selected learner characteristics	Stored profile for conversion and style explanation
2 Load prediction	Interpret changing video behavior	Accepted or skipped window; load for conversion and support
3 Resource conversion	Prepare and transform educator material	Source knowledge and adapted content using Components 1 and 2
4 Explainable supports	Turn evidence into understandable action	Explanations, reviewed guidance, next lesson plan, and MCQs

The boundaries are complementary. Profiling supplies context but does not estimate each learning segment's load. Load prediction supplies a changing state but does not determine which content facts are valid. Resource processing defines the source boundary but does not validate a classifier. Explanation and recommendation use all three inputs while maintaining separate review policies. This division makes failures easier to locate and avoids attributing the whole platform's performance to one module.

2.4 Overall aim and group objectives

The overall aim is to develop and evaluate an integrated adaptive learning platform that connects learner profiles, behavioral cognitive-load estimates, and educator resources to interpretable and reviewable educational support.

- **O1:** Construct a multi-dimensional learner profile while preserving the meaning and limitations of each assessment.
- **O2:** Estimate five cognitive-load levels from consistent two-minute behavioral windows, with a clear response when evidence is insufficient.
- **O3:** Prepare source-linked lesson knowledge and generate explanations conditioned on learner profiles and cognitive load.
- **O4:** Connect LIME and SHAP evidence to teacher-readable explanations, constrained learner guidance, and course-level recommendations.
- **O5:** Generate and mark lecture-grounded MCQs with structural validation.
- **O6:** Verify component interfaces and evaluate technical behavior and preliminary stakeholder responses without overstating learning effectiveness.

Each objective is associated with an output that can be checked: a stored profile, a prediction record, prepared resource knowledge, a reviewed guidance record, a validated quiz, or an evaluation result. Chapter 6 maps these outputs to evidence and remaining work. This prevents objectives concerning educational outcomes from being marked completely solely because a service endpoint exists.

2.5 Scope assumptions and constraints

The platform supports teacher, student, and administrator workflows in a browser. Teachers organize courses into sections and subsections and upload authorized

resources. Students complete profiling activities, enroll in prepared courses, watch lessons, request personalized explanations, and complete quizzes. The implementation combines separate services and databases connected through a gateway. Availability depends on configured databases, asset storage, and external model providers.

The load model uses six video/content features; stored quiz and adaptation metrics are not current classifier inputs. Profiling uses approximate webcam-derived gaze observations rather than dedicated eye-tracking hardware. Resource conversion assumes usable uploaded material and cannot guarantee faithful recovery of every equation, scanned page, or diagram. The fifteen-minute video ceiling is an engineering choice rather than a demonstrated optimal learning duration.

The evaluation is bounded by its data. The undergraduate load study, university profiling records, teacher comparisons, and Grade 8 classroom pilot represent different populations and activities. No pooled participant total is used. Formal ethics approval measured commercial demand, general accessibility compliance, and controlled learning gains are not established by the current results. Missing model and dataset provenance remains a reproducibility constraint.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

Lumora connects an educator-side preparation process with a learner-side interaction and support process. Profile information is established through initial activities. Prepared course knowledge becomes available after upload processing. During viewing, a two-minute observation window produces a load estimate when the reliability checks pass. The learner profile and current load then condition the presentation of the educator's knowledge.

Two explanation branches support teachers. Cognitive-style explanations describe influential observations behind a selected profile class. Cognitive-load explanations identify behaviors supporting a lesson-level load estimate and inform reviewed learner guidance. Course-level summaries support next-lesson planning, while extracted lecture text supplies the factual context for assessment. These outputs share identifiers and evidence but have different purposes and validation requirements.

3.2 High level architecture

The React interface communicates with an Express API gateway. The gateway routes identity and enrolment requests to the core backend, course and upload requests to the resource service, behavioral predictions to FastAPI, and support requests to the corresponding AI services. MongoDB holds identity, course structure, resource metadata, and profiling observations. MySQL holds load windows, explanations, quizzes, and recommendation records. *Figure 1* shows these boundaries and the shared infrastructure.

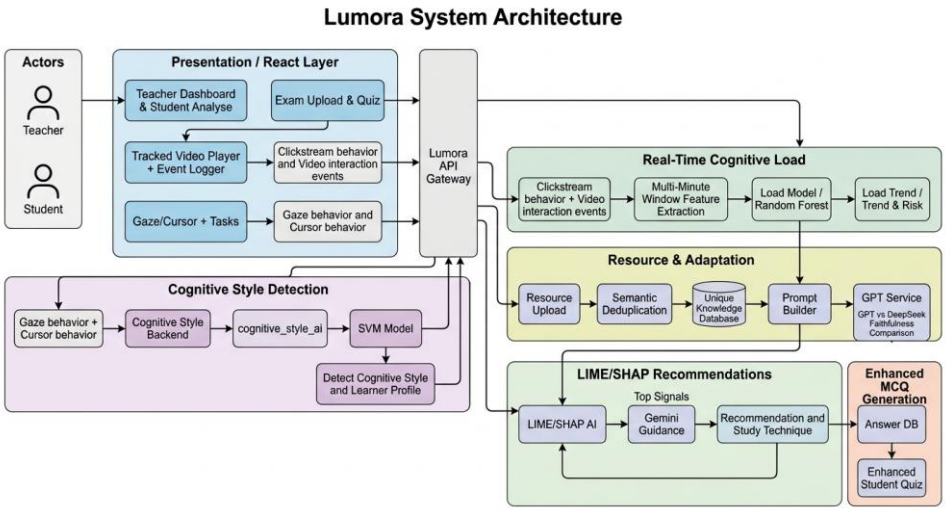


Figure 1 Integrated platform architecture and service boundaries

The research components span more than four executable services. Component 3 uses the resource service, the Hugging Face generation service, and the DeepSeek service. Component 4 uses LIME, SHAP, style explainability, recommendation, and examination services. The gateway is a routing boundary; it does not make every downstream route uniformly authorised. Identity, ownership, and enrolment checks must remain explicit at the services that expose protected information.

3.3 Component 1 - Learner profile and cognitive style detection

The first component, led by Imesh L A R (IT22281364), combines learning-approach and cognitive-style information. ASSIST responses contribute a Deep, Strategic, or Surface learning approach. The analytic–wholistic workflow combines questionnaire and task-related information. A split-screen learning activity presents visual-supported and text-only material and records cursor activity, independent scrolling, approximate gaze distribution, and first interaction preference.

The profiling study records 212 initial responses. Data preparation excludes four records with missing required values and ten with gaze duration below 10,000 ms. The dataset summary identifies 198 prepared records, split into 158 training and 40 testing records. However, the exclusion groups require an explicit overlap check to reproduce the final count. The classification experiment describes a Support Vector Machine with an RBF kernel, standardization, balanced weights, and training-only synthetic

minority oversampling. Support-vector classification and SMOTE are established methods [8] [9].

The operational style-explanation path currently loads a separate three-class Random Forest artifact using four features: image cursor ratio, image scroll ratio, image gaze ratio, and first interaction with visual content. Its labels are Visual, Verbal, and Intermediate. The SVM experiment and the deployed Random Forest artifact are distinct versions; numerical performance cannot be transferred between them. No complete numerical style-classification result is established here. The contribution is the structured profiling and behavioral collection workflow, with model-version reconciliation remaining necessary.

3.4 Component 2 - Real time cognitive load prediction

The second component, led by Subasinghe M.H.R.I. (IT22089786), estimates learning effort from video behavior. Its six inputs are pause frequency, forward navigation, rewatch count, playback-rate changes, idle duration, and time on content. The service uses two-minute windows and associates each observation with student, lesson, session, and window identifiers.

The study involved 171 undergraduate computing students: 108 laboratory and 63 remote participants. It retained 1,420 valid windows from two instructional videos. After each segment, learners supplied a nine-point Paas mental-effort rating [10]. Ratings 1–2, 3–4, 5, 6–7, and 8–9 were mapped respectively to Very Low, Low, Medium, High, and Very High. The mapping is the project's target definition, not a universal five-level interpretation of the instrument.

An 80/20 participant-level split kept each learner's windows in one partition. The held-out evaluation contained 293 windows. Random Forest [11] achieved 79.52% accuracy and 79.55% macro F1, outperforming the other evaluated candidates. Deployment combines median imputation with 200 trees, maximum depth 16, minimum split size 10, balanced class weights, and random state 42. Fixed reliability rules skip windows with insufficient watch time or prolonged inactivity. Saved accepted predictions support explanation dispatch, trend history, and adaptation.

3.5 Component 3 - Resource upload and adaptive content conversion

The third component, led by Jayasinghe J. M. S. N (IT22166074), connects uploaded resources to learner-specific presentations. Course subsections hold a required video and optional presentation, PDF, and images. Assets are stored externally, while MongoDB preserves their associations and preparation status. A background worker transcribes speech, extracts document text and images, processes equations where enabled, and assembles deduplicated lesson knowledge.

Sentence-BERT supplies the methodological background for semantic comparison [12], while MiniLM supports compact transformer representations [13]. The implementation uses all-MiniLM-L6-v2 when available and lexical fallbacks otherwise. Presentation material has priority, followed by PDF additions and transcript additions. Knowledge assembly remains extractive, avoiding a second generative summary before adaptation. Detected mathematical blocks are protected, but protection preserves the extracted representation even when that representation is wrong.

The pedagogical prompt combines knowledge, learner profile, presentation-related style, analytic or holistic presentation, and current load. Hugging Face and DeepSeek generate alternatives. A selector combines a source-similarity proxy with readability matching using weights of 0.65 and 0.35. These weights are configured heuristics rather than experimentally validated optimal values. Offline verification passed 18 of 19 focused checks and exposed incomplete mathematical symbol preservation in a PDF fixture. This establishes selected technical behavior without establishing hallucination elimination or learning improvement.

3.6 Component 4 - Explainable support recommendations and assessment

The fourth component, led by Dilshan W.G.W (IT22055712), turns model evidence and lecture content into task-specific educational support. Local Interpretable Model-agnostic Explanations (LIME) approximate local model behavior [14]. SHapley Additive exPlanations (SHAP) allocate additive feature contributions [15]. The platform retains raw attribution evidence while producing teacher-oriented language.

Style analysis ranks the mean of separately normalized absolute LIME and SHAP contributions and preserves direction. It generates a concise explanation of the predicted style without teaching advice. Load guidance uses a different policy: positive contributions are normalized within each explainer and combined with selecting up to three signals supporting elevated load. This selected list does not represent every influence on prediction. Structured generation provides a teacher explanation, four learner actions, and one or two techniques from a fixed catalogue. Feedback research and retrieval practice motivates actionable support and self-check activities [16] [17].

Next-lesson recommendations combine load distributions, recurring signals, and available style summaries. Chain-of-Verification structures drafting, verification questions, evidence-based answers, and final revision [18]. A deterministic fallback supports continuity when verification fails. The examination path generates exactly ten lecture-grounded MCQs with four options and retains answer keys server-side during an attempt. Teacher review applies to guidance and next-lesson recommendations; mandatory approval before every learner quiz is not enforced by the current MCQ workflow.

3.7 End to end user and data flow

Figure 2 presents the integrated flow in the four-component order. A student first establishes profile information. When a prepared course is available, the student enrolls and watches a lesson. The interface captures video interactions and requests a prediction after each complete window. Accepted predictions are stored, used for trend display, and supplied to explanation and conversion. A skipped response communicates inadequate evidence and does not become a valid load class.

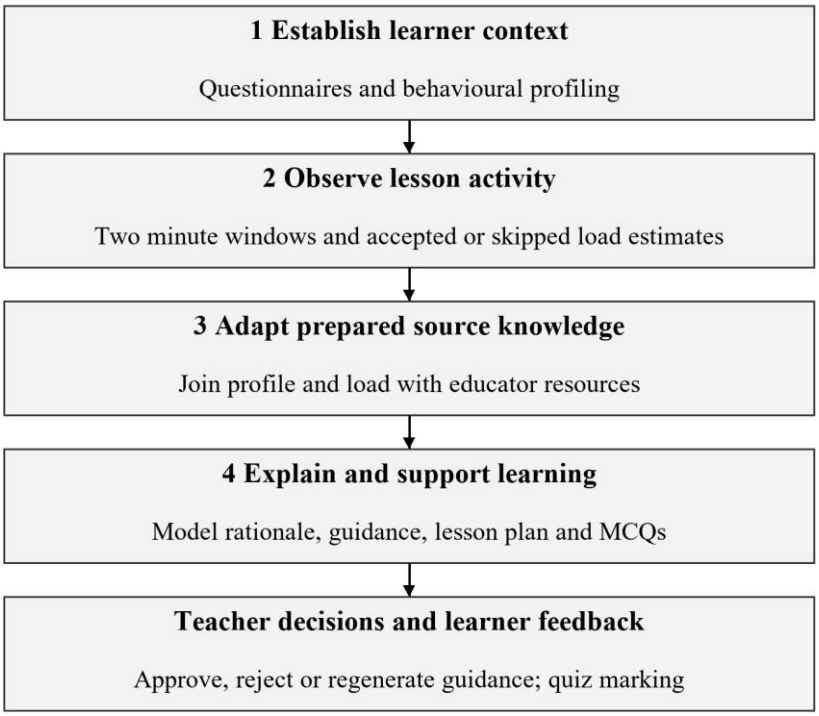


Figure 2 Four component system flow and feedback

Resource preparation usually occurs before viewing and is reused by many learners. Its position as Component 3 describes the research responsibility and adaptation stage, rather than requiring a teacher to upload files after a student is assessed. At conversion time, Component 3 joins prepared content with outputs from Components 1 and 2. Component 4 subsequently explains model evidence and creates reviewable guidance or assessment. Teacher decisions and learner feedback close the support workflow, although feedback-driven model retraining is not established as an automatic operation.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The integration strategy separates data capture, inference, content preparation, and support generation while exposing one browser-facing gateway. JSON over HTTP allows Python model services and Node.js application services to exchange data without sharing process memory. Shared identifiers link records across the boundaries. This modular design supports independent development but also makes consistent contracts and failure interpretation essential.

Integration follows dependency order. Identity and course records establish the context. Profile data and prepared knowledge supply stable inputs. Session windows supply dynamic load evidence. Explanation and generation use those inputs and write their own outputs. Review actions then determine whether learner guidance is shared. The approach avoids using a generated recommendation as if it were an observed interaction or a classifier label.

Static interfaces, automated function tests, and recorded demonstrations establish different levels of integration evidence. They do not collectively substitute for a live regression test spanning browser event, databases, external asset storage, both candidate providers, and teacher review. A release should preserve a known dataset, model signatures, service configuration, and a complete scenario trace so that an unexpected result can be reproduced across services.

4.2 Interfaces data contracts and shared models

The central contract links student_id, lesson_id, session_id, minute_index, window_start, and window_end to a feature snapshot. The prediction model receives the six features in a fixed order. Reliability responses distinguish reliable from not_reliable; skipped results contain a reason, null predicted class, and zero confidence. *Table 2* summarizes the principal exchanged records. Appendix A specifies representative endpoint responsibilities.

Table 2 Principal shared records

Record	Key fields	Meaning
Learner profile	User ID; approach; style labels	Contextual profile, separate from current load
Feature window	Student, lesson, session, index; six features	Behavior over one completed observation window
Prediction	Status; label; score; confidence; time	Model estimate or explicit insufficient evidence
Prepared knowledge	Course, section, subsection; source URLs; extracts	Reused instructional evidence and preparation state
Explanation	Prediction or profile ID; LIME; SHAP; signals	Local model evidence and generated rationale
Review and feedback	Review state; decision; technique feedback	Teacher action and subsequent learner response
Quiz	Lesson context; items; options; stored key	Lecture grounded assessment with separate marking

Identifier meaning needs particular attention. In the current tracked course flow, the course identifier is passed as lesson_id. Profile analysis is student-level, with a lesson recorded mainly as creation context. Recommendation code derives the relevant course lesson identifiers from stored structure. These are implemented conventions rather than a uniform semantic definition of lesson throughout the project. A shared identifier specification is needed before combining subsection-specific and course-level histories.

The current schemes validate types and required fields but are not a versioned cross-service contract registry. Model feature order, label aliases, and probability interpretation must be checked independently of API shape. UTC creation times and ISO-formatted window times support tracing. Parameterized SQL separates values from queries, while MongoDB relationships and cross-database links remain largely application-managed. Concurrent updates and partial writes therefore require explicit handling.

4.3 Shared nonfunctional requirements

Reliability requires correct interpretation of missing data and dependency failure. The prediction gate should avoid assigning a low-load category to inactivity. Content preparation should distinguish completed processing from verified extraction. Generated guidance should remain pending until approved, and rejected guidance should not appear in the student's shared list. These requirements concern the trustworthiness of workflow states as much as service availability.

Security and privacy require authenticated access, role and ownership checks, protected configuration, evidence minimization, and appropriate retention. The current system contains access controls in several workflows. Production readiness requires eliminating fallback credentials, reviewing direct-access routes, and testing authorization for student records and uploaded material. UNESCO's guidance emphasizes human agency, privacy, and careful institutional use of generative AI [19].

Performance must be measured across the whole request. The local 80 ms prediction average include every production dependency or the two-minute observation period. Transcription, extraction, attribution, and generation can take longer and should expose useful progress states. Maintainability requires pinned model dependencies, versioned schemas, reusable client configuration, and service-level diagnostics. Accessibility requires keyboard use, readable diagrams and equations, meaningful error messages, and an alternative to webcam-dependent profiling. These qualities remain targets for broader validation.

4.4 Integrated workflow and exception handling

An upload request persists assets and subsection metadata before background preparation begins. The subsection progresses through queued, processing, and ready or failed states. On restart, persisted queued and interrupted records can be rescheduled. The main queue is for an in-memory FIFO worker within one Node.js process; it does not provide distributed job ownership. A separately scheduled rebuild path also needs coordination before multi-instance deployment.

A valid prediction request passes the reliability gate, executes inference, and optionally saves a window and prediction log. It then attempts LIME dispatch. Dispatch records

reduce repeated successful submissions. The raw-event automation path also forwards completed feature windows to LIME, but this is not identical to creating the prediction-log records used by trend analysis. The two routes must be tested for consistent feature definitions and duplicate handling.

Table 3 summarizes important exceptions and their actual responses. *Figure 3* shows the content-preparation lifecycle and where quality checks remain separate from readiness. A Whisper failure can leave usable document material; one generation provider can supply the only candidate; failed neural similarity can trigger lexical comparison. Each case reduces capability and should remain visible. Neither a fallback nor a ready flag certifies instructional correctness.

Table 3 Failure handling across the integrated workflow

Condition	Current response
Insufficient viewing	Skip prediction and show reason
MySQL unavailable	Direct inference can still run
CSV locked	Append to fallback CSV
Whisper failure	Continue if documents are usable
Interrupted preparation	Reschedule queued or processing records
Neural similarity unavailable	Lexical scoring fallback
Only one candidate succeeds	Return available candidate
Style explanation generation fails	Return controlled failure for retry
Recommendation verification fails	Deterministic fallback
Malformed MCQ output	Reject structural violations

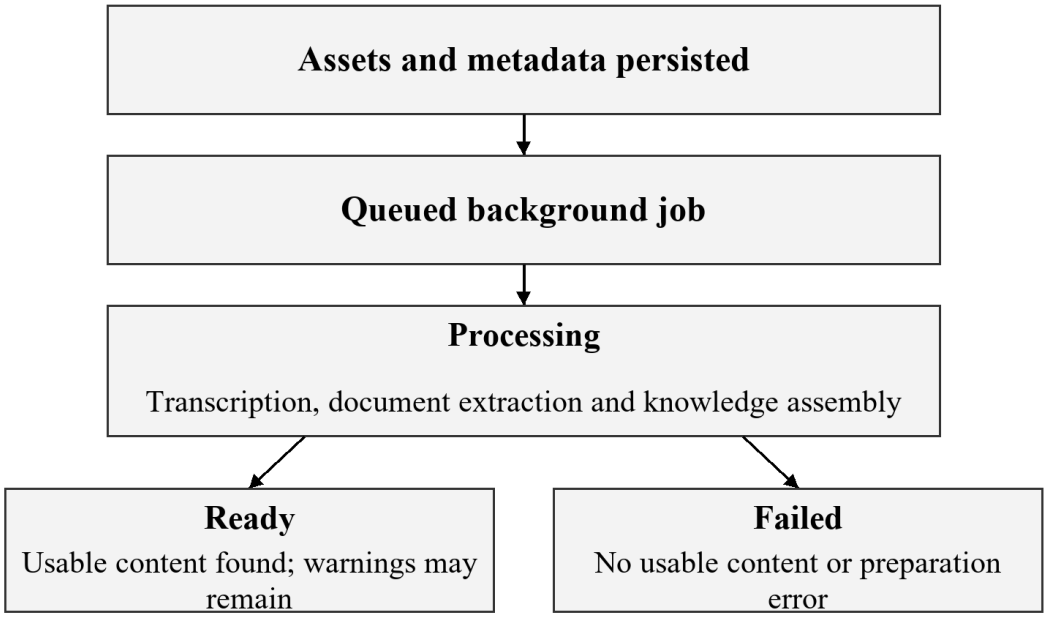


Figure 3 Resource preparation states and recovery

4.5 Team contribution and dependency matrix

The four responsibilities are assigned in system-flow order. *Table 4* identifies the named lead, the principal research and development work, and the required handoff. Shared execution includes frontend integration, identity propagation, course context, interface alignment, and testing of cross-component scenarios. These shared activities are necessary because a correct isolated module can still receive the wrong learner or resource record.

Table 4 Team responsibilities and dependencies

Member and ID	Primary responsibility	Integration handoff
Imesh L A R IT22281364	Profiling instruments, behavioral collection and classification workflow	Student profile and style model inputs
Subasinghe M.H.R.I IT22089786	Video feature collection, load model, reliability and trends	Saved load windows and learner state
Jayasinghe J. M. S. N IT22166074	Resource upload, extraction, knowledge assembly and conversion	Prepared source knowledge and generated candidates

Member and ID	Primary responsibility	Integration handoff
Dilshan W.G.W IT22055712	Explanations, reviewed guidance, lesson recommendation and MCQs	Evidence translation, review states and assessment outputs

The matrix records responsibility boundaries rather than assigning every source-code line to one person. Component 1 owns profiling research and collection; Component 2 owns load modelling and its service; Component 3 owns resource preparation and conversion; Component 4 owns explanatory support, recommendations, and assessment. Approval and signatures for the contribution declaration remain separate from this technical description. Appendix B provides an unsigned confirmation record for the four members.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The frontend uses React and Vite, with React Router for navigation, Axios for API access, and charting libraries for analytical views. Markdown, Mermaid, and KaTeX support formatted explanations, diagrams, and equations. MediaPipe supports facial and iris-related landmarks during approximate gaze estimation. These browser functions connect the student learning experience with profiling and prediction without specialized eye-tracking devices.

Node.js and Express implement the gateway, core application backend, resource workflow, assistants, recommendations, and exam service. FastAPI and Pydantic implement cognitive-load and explainability interfaces. Scikit-learn, pandas, NumPy, joblib, LIME, and SHAP support inference and attribution. MongoDB stores flexible application and interaction records, while MySQL and SQLAlchemy support structured analytical persistence. Cloudinary, Whisper, PDF/PPTX extraction helpers, and external language-model endpoints supply preparation and generation capabilities. *Table 5* summarizes the shared stack by role.

Table 5 Shared technology stack

Layer	Technologies	Purpose
Browser	React, Vite, Axios, React Router	Course, profile and analysis workflows
Presentation	Markdown, Mermaid, KaTeX, charts	Text, equations, diagrams and summaries
Application services	Node.js, Express, JWT, Multer	Gateway, identity, courses and uploads
Model services	FastAPI, Pydantic, scikit learn	Validated inference and AI interfaces
Explanation	LIME, SHAP, Gemini	Attribution and teacher readable text
Preparation	Whisper, PDF/PPTX helpers, MiniLM	Speech, document and knowledge processing

Layer	Technologies	Purpose
Persistence	MongoDB, MySQL, SQLAlchemy	Application records and analytical data
External assets and generation	Cloudinary, Hugging Face, DeepSeek	Media storage and candidate content

The split between documents and relational databases reflects service-specific records. It also introduces reconciliation work: there is no single transaction spanning course metadata, behavioral windows, explanation rows, and quiz records. Data ownership should therefore remain explicit, and derived summaries should be traceable to their input records. The older explainable-analysis path uses a separate database structure and should not be assumed to read the current LIME prediction tables automatically.

5.2 Development and integration process

Development produces reusable components around a shared React application and gateway. The practical integration work includes aligning route prefixes, propagating token and ownership information, storing model-ready records, and connecting UI states to service responses. Component tests isolate deterministic behavior through controlled inputs and simulated provider responses. Frontend demonstrations support inspection of visible predictions, features, skipped windows, and recent trends.

Version control contains the implementation and multiple development branches, while service-specific manifests and Python requirements describe dependencies. Research artifacts include model files, schemas, verification scripts, and evaluation records. A reproducible release nevertheless requires more than source availability: the exact dataset snapshot, participant split, fitted preprocessing, provider revision, and deployment settings must be archived together.

The current evidence does not establish a complete continuous-integration pipeline, formal sprint history, or an automated cross-service release process. These practices should be introduced as improvements rather than retroactively claimed as completed work. The recommended development sequence is to define interface expectations, verify producer and consumer behavior with the same fixture, run relevant regressions,

and then execute a live scenario with persisted evidence. This sequence is especially important when changing model labels, feature calculations, or resource extraction.

5.3 Deployment and operational configuration

The local gateway defaults to port 4000, with the frontend normally on 5173. The main application backend uses 5001; resource upload uses 5000; Hugging Face generation uses 5002; DeepSeek uses 5004; next-lesson recommendations use 5003; load inference uses 8021; LIME uses 8110; SHAP uses 8111; style explainability uses 8112; and the cognitive-style backend is expected on 8003. Exam requests pass through the core backend to the exam service. Environment configuration should be treated as authoritative for a particular deployment rather than copying inconsistent example ports.

The profiling collection application used Azure Container Apps, and the classroom pilot used a hosted learning workflow. Those activities demonstrate specific hosted use, but the classroom record does not identify a build revision or exact provider configuration. The repository also lacks a complete, verified deployment package for every service. Local launch commands, health endpoints, and configuration variables support operation, while full deployment reproducibility remains incomplete.

Operational configuration must keep secrets outside the report and source control. Databases should start before services that initialize tables or depend on stored records. Health checks should separately expose databases and model status. Backups need to cover application metadata, analytical records, and uploaded assets, with a tested restoration sequence. A future deployment should include schema migrations, central logging, job supervision, request limits, and a trace identifier connecting browser actions to stored outputs. These are readiness requirements, not completed benchmark findings.

5.4 Evidence of team execution

The integrated artifacts establish a shared project with clearly connected responsibilities. Profiling activities populate learner context; the load API receives interaction features from the course view; the resource worker prepares knowledge; prompt construction consumes profile and load values; and explanation and

recommendation interfaces expose the resulting records to teachers. Course-linked assessment connects extracted teaching content to generated questions and marking.

Technical execution is also evidenced by the separate verification groups: prediction-service tests, resource-helper checks, guidance and signal tests, recommendation verification tests, MCQ validation tests, and style-service tests. These groups cover different components and should retain their original scope. A total obtained by adding every count would be misleading where revisions, fixtures, or reported groups overlap.

The team contribution record identifies responsibility, but exact commit attribution, meeting attendance, planner activity, and review dates are not reconstructed from assumptions. A final research archive should retain a release tag, contribution confirmations, test logs, and the evaluation instruments. Preserving these records would make both the engineering process and the experimental conclusions easier to audit without expanding the common report into raw activity logs.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

Evaluation addresses five questions: whether learner information can be captured and represented; whether video behavior predicts the grouped mental-effort target; whether source preparation preserves usable content; whether generated support is structurally controlled and reviewable; and how teachers and learners perceive selected outputs. The evaluation distinguishes function correctness, model validity, content integrity, and educational usefulness.

For load classification, the measures include accuracy, macro precision, macro recall, macro F1, balanced accuracy, weighted F1, and class-level recall. Participant separation is essential because repeated windows from one learner are not independent participants. For profiling, training-only standardization and resampling preserve the test boundary, while the deployed model version needs separate validation. For resource processing, successful extraction, mathematical fidelity, duplicate removal, and candidate selection are distinct outcomes.

For generated support, acceptance checks include permitted techniques, observed behavior matching, review-state handling, verification evidence paths, exact MCQ count, option count, answer labels, and duplication. Human evaluation assesses clarity, relevance, usefulness, and suitability. These checks can establish useful function behavior without proving semantic correctness. Table 6 records the evaluation contexts and their interpretation boundaries. The success criterion is evidence appropriate to each claim; there is no defensible single metric for the whole system.

Table 6 Evaluation contexts and evidence boundaries

Evaluation	Sample or material	Interpretation
Profiling collection	212 initial records; 198 listed after cleaning	Collection and preparation; numeric model result incomplete
Load classification	171 learners; 1,420 windows; 293 test windows	Agreement with grouped Paas ratings

Evaluation	Sample or material	Interpretation
Resource verification	19 focused checks; mathematical fixtures	Helper behavior and a retained fidelity failure
Support automation	39 reported tests plus separate 10 style tests	Specified functions, with controlled dependencies
Guidance comparison	3 teachers; 3 scenarios; 9 paired cases	Descriptive expert preference
MCQ comparison	3 teachers; 30 item judgments per method	Suitability and answer key review
Classroom pilot	10 Grade 8 learners; 1 teacher; one lesson	Single post lesson scores and clarity ratings

6.2 Integration and end to end testing

The cognitive-load component records 15 passing automated tests covering feature extraction, idle intervals, completed-window dispatch, duplicate handling, output normalization, reliability rules, storage conversion, and trend analysis. The resource component records thirteen passing JavaScript checks and five of six Python checks. Its one failed assertion concerns mathematical symbol fidelity in a PDF. That failure remains part of the result rather than being converted into a successful availability check.

The support test summary records 39 passing checks: twenty for individual guidance and signal aggregation, thirteen for next-lesson and verification logic, four for MCQ generation, and two for lecture PDF extraction. A separate style-explanation suite records ten passing tests for attribution aggregation, teacher-oriented prompts, label handling, caching, model refresh, MongoDB synchronization, and artifact loading. *Table 7* retains these counts by verification group.

Table 7 Recorded automated verification by group

Group	Outcome	Scope
Load prediction	15/15 passed	Extraction, reliability, dispatch and trends

Group	Outcome	Scope
Resource JavaScript	13/13 passed	Extraction helpers, prompts, scoring and fallbacks
Resource Python	5/6 passed	PDF symbol fidelity failed; other fixtures passed
Guidance and signals	20/20 passed	Ranking, persistence, reviews and validation
Next lesson verification	13/13 passed	Evidence checks, recommendation and fallback
MCQ generation	4/4 passed	Schema and generation handling
Lecture PDF extraction	2/2 passed	Exam extraction scenarios
Style explanation	10/10 passed	Aggregation, prompts, sync, cache and model loading

These automated tests use controlled dependencies where appropriate. They do not establish live model-provider quality, concurrent-load behavior, extraction correctness for an unrestricted corpus, or production authorization. The classroom pilot provides an example of hosted lesson and quiz use, but it does not exercise every component or failure route. A complete live regression should follow one synthetic learner through profile creation, resource preparation, a reliable and skipped window, stored explanations, teacher sharing, feedback, and quiz submission. That consolidated release-level evidence remains necessary.

6.3 Overall results

Cognitive Style Classification Results

The Support Vector Machine (SVM) classifier with a Radial Basis Function (RBF) kernel achieved the strongest performance among the compared algorithms for predicting visual–verbal cognitive style. As shown in Table 8, the SVM model achieved an accuracy of 77.50% and a weighted F1-score of 0.7698. K-Nearest Neighbors (KNN) achieved the second-highest accuracy of 72.50%, followed by Random Forest at 65.00%, Decision Tree at 60.00%, and Logistic Regression at

57.50%. These results support the selection of SVM as the final classifier among the evaluated models under the study conditions.

Table 8 Cognitive Style classifier comparison

Algorithm	Accuracy
Support Vector Machine (RBF)	77.50
K-Nearest Neighbors (KNN)	72.50
K-Nearest Neighbors (KNN)	72.50
Random Forest	65.00
Decision Tree	60.00
Logistic Regression	57.50

The class-wise evaluation indicates differences in classification performance across the cognitive style categories. The model achieved an overall accuracy of 77.50%, with a macro-average precision of 0.7645, macro-average recall of 0.7750, and macro-average F1-score of 0.7698. The weighted-average precision, recall, and F1-score were 0.7698, 0.7875, and 0.7750, respectively, according to the supplied evaluation summary. These metrics provide complementary views of model performance across the evaluated classes.

Table 9 Class-wise cognitive style classification performance

Class	Precision	Recall	F1-score	Support
Moderate/Intermediate	0.8108	0.7500	0.8824	17
Verbal	0.7826	0.7500	0.8182	11
Visual	0.7000	0.8750	0.5833	12
Macro average	0.7645	0.7750	0.7613	40
Weighted average	0.7698	0.7875	0.7750	40
Accuracy	—	—	0.7750	40

The results suggest that the SVM classifier can identify cognitive style categories from the selected behavioral features, although its performance differs between classes. Therefore, overall accuracy should not be interpreted as evidence that every category is predicted equally well. The results are limited to the available dataset and evaluation procedure and do not establish that cognitive styles are permanent characteristics or that the model will achieve the same performance with other learners.

Cognitive Load Classification Results

The load classifier achieved the strongest performance among the compared algorithms. *Table 10* reports the comparison on the stated evaluation, and *Figure 4* compares accuracy and macro F1. Random Forest reached 79.52% accuracy, corresponding to 233 correct predictions among 293 test windows. Its balanced accuracy and macro recall were 79.48%, macro precision was 79.87%, and weighted F1 was 79.56%. These values describe agreement with grouped self-reported effort within the study conditions.

Table 10 Cognitive load classifier comparison in percentage units

Model	Accuracy	Macro precision	Macro recall	Macro F1
Logistic Regression	66.89	66.02	66.83	66.23
Decision Tree	67.58	66.98	67.58	67.15
XGBoost	75.77	76.50	75.71	75.95
Random Forest	79.52	79.87	79.48	79.55

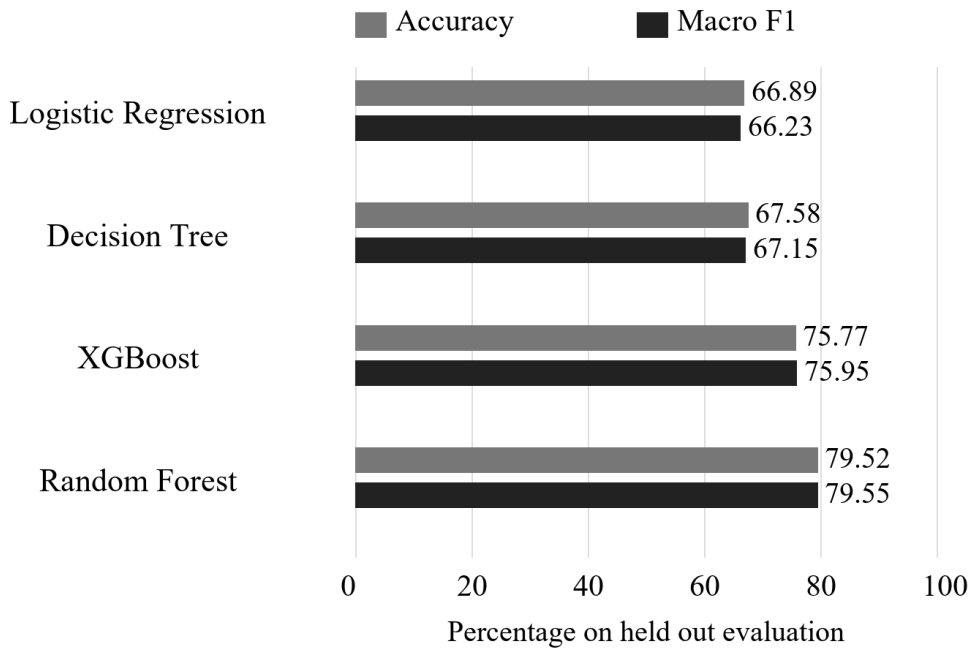


Figure 4 Cognitive load classifier accuracy and macro F1

Class-level performance was uneven. Very Low recall was 95.00%; Low and High each achieved 69.49%; Medium achieved 84.48%; and Very High achieved 78.95%. *Table 11* preserves the supports and correct counts. The lower recall for adjacent intermediate categories matters because a useful overall score can conceal uncertainty in decisions that trigger adaptation. Feature importance ranked idle duration first, followed by pauses, time on content, navigation, rewatches, and rate changes. These associations describe the fitted classifier and are not causal explanations of mental effort.

Table 11 Cognitive load class recall on held out windows

Class	Support	Correct	Recall (%)
Very Low	60	57	95.00
Low	59	41	69.49
Medium	58	49	84.48
High	59	41	69.49
Very High	57	45	78.95

Class	Support	Correct	Recall (%)
Total	293	233	Accuracy 79.52

The local benchmark recorded 800 ms across ten requests, averaging 80 ms per request. This is a small local responsiveness observation, not a production latency guarantee. Predictions become available after a complete two-minute window; that interval is separate from inference processing time. Database persistence and synchronous downstream dispatch can also affect the saved-prediction response.

Profiling has collection and preprocessing evidence, including 212 initial and 198 listed prepared records, but incomplete numerical classification reporting and model-version differences constrain a final accuracy claim. Resource checks returned five equation strings from the presentation fixture and eight mathematical candidates from the PDF path. The PDF symbol-preservation assertion failed. Thus, a returned expression cannot be assumed mathematically faithful. The authored candidate-selection fixture favored source alignment under lexical comparison, but it did not compare actual generated candidates or estimate a hallucination rate.

6.4 Stakeholder and user validation

Three ICT teachers evaluated baseline and system guidance across three scenarios, producing nine paired comparisons. The system was preferred in all nine. Across five rating criteria, the system mean was 4.84/5 against 3.87/5 for the baseline. The overall-usefulness criterion was 5.00/5 against 4.00/5. *Table 12* and *Figure 5* present the descriptive criterion means. The nine cases are clustered within three evaluators and are not nine independent teachers.

Table 12 Teacher guidance comparison across nine paired cases

Criterion	Baseline (/5)	System (/5)
Clarity	3.44	4.89
Evidence relevance	3.67	4.78
Recommendation usefulness	4.33	4.78

Criterion	Baseline (/5)	System (/5)
Technique suitability	3.89	4.78
Overall usefulness	4.00	5.00
Five criterion mean	3.87	4.84

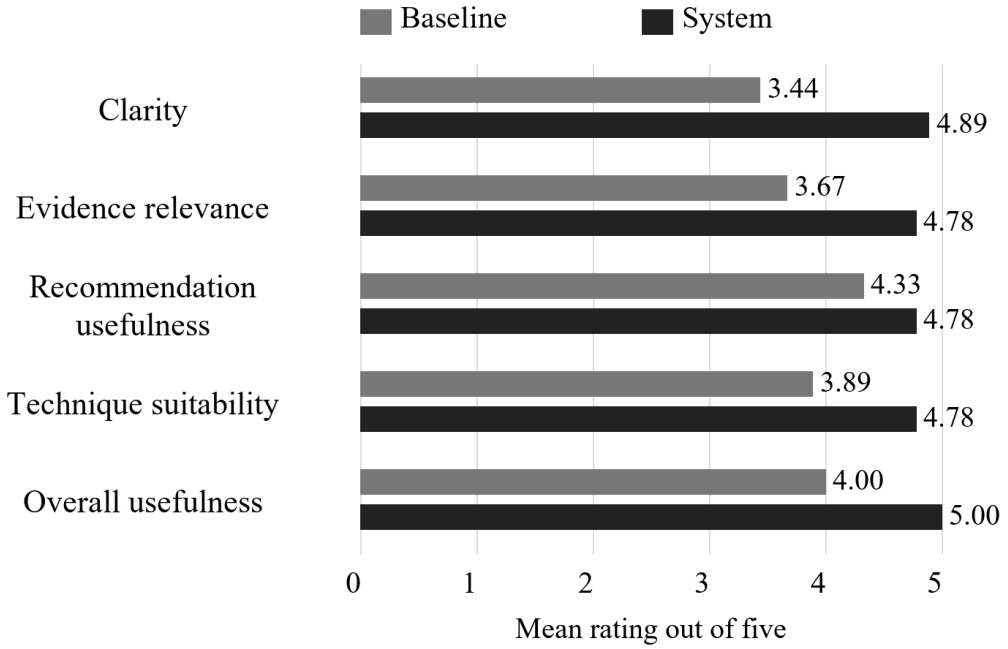


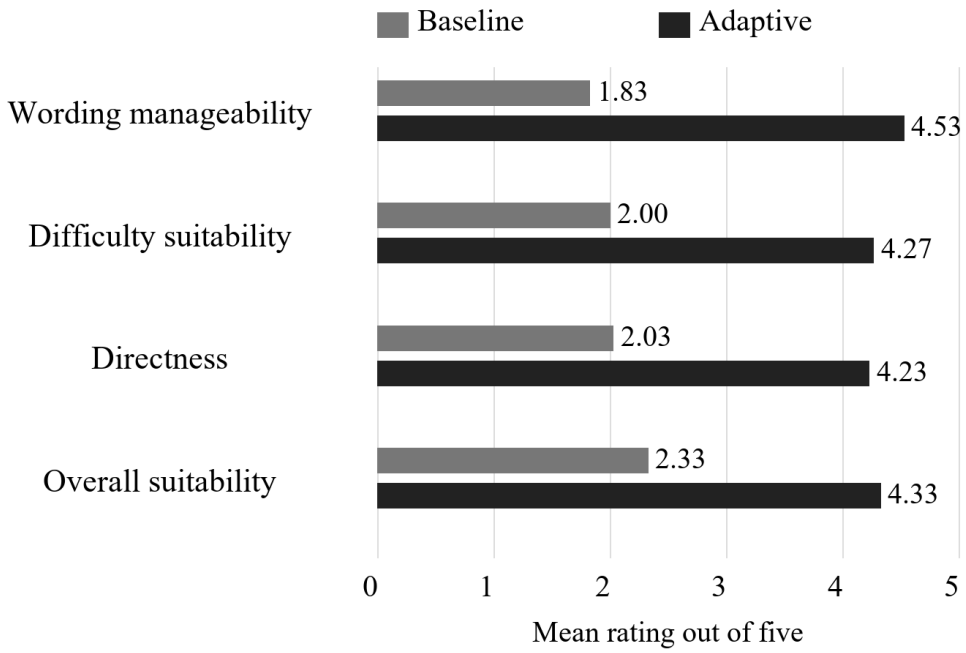
Figure 5 Teacher guidance criterion means across nine paired cases

For the MCQ comparison, three teachers evaluated baseline and adaptive sets. Adaptive mean ratings were higher for wording manageability, difficulty suitability, directness, and overall suitability, as shown in *Table 13*. All three preferred the adaptive set. Answer-key review recorded affirmative correct-and-unique judgments for 30/30 baseline items and 29/30 adaptive items. The remaining adaptive judgment lacked an explanatory note and requires review; it is not independently established as an answer error. *Figure 6* contrasts preference-related ratings with answer-key checks.

Table 13 Descriptive expert ratings of MCQ sets

Criterion	Baseline (/5)	Adaptive (/5)
Wording manageability	1.83	4.53
Difficulty suitability	2.00	4.27

Criterion	Baseline (/5)	Adaptive (/5)
Directness	2.03	4.23
Overall suitability	2.33	4.33



Correct and unique answer judgments: baseline 30 of 30; adaptive 29 of 30.
One adaptive judgment requires review.

Figure 6 Adaptive MCQ ratings and answer key review boundary

The exploratory classroom session took place on 3 October 2026 with ten Grade 8 learners and one teacher, using the INTERNET lesson. Every learner completed ten generated questions. Correct counts were 8, 7, 9, 9, 8, 7, 8, 7, 7, and 8, totalling 78/100 responses, or 78.0%. The mean score was 7.8/10. MCQ clarity ratings averaged 4.30/5. Displayed load categories were Low for one learner, Medium for two, High for six, and Very High for one. *Table 14* summarizes the session; anonymised records appear in Appendix C.

Table 14 Exploratory classroom pilot summary

Measure	Result
Session	3 October 2026; INTERNET lesson

Measure	Result
Participants	10 Grade 8 learners and 1 teacher
Questions per learner	10
Correct responses	78 of 100 (78.0%)
Mean learner score	7.8 of 10
Mean MCQ clarity	4.30 of 5; 10 responses
Displayed load	Low 1; Medium 2; High 6; Very High 1
Teacher review	1 completed review; recommendation appropriate
Design limitation	No pre test, comparison condition or difficulty ratings

The single teacher marked the content alignment and answer keys positively, judged the difficulty very appropriate and wording clear, and considered the next-lesson recommendation appropriate. This is one person's review of one session. Exact learner scale anchors, perceived-difficulty responses, and written learner comments were not recorded. There was no pre-test or comparison group. The scores therefore describe an attempt rather than a learning gain. The teacher comparisons did not separately evaluate cognitive-style explanations, and their ratings cannot be assigned to that output.

6.5 Achievement of common objectives

Table 15 maps each group objective to the available evidence and remaining work. The project establishes substantial implementation of profiling, load prediction, resource preparation, explainable support, and assessment. It also exposes meaningful limits that change the interpretation of completion: model deployment must match evaluated artifacts, content readiness must preserve extraction warnings, and assessment structure must remain separate from answer correctness.

Table 15 Common objective achievement and remaining validation

Objective	Evidence	Status and remaining work
O1 Learner profile	Capture, instruments and prepared records	Implemented; reconcile model versions and numeric evaluation
O2 Load estimation	Five class model, 79.52% accuracy, reliability gate	Implemented and evaluated within study; align feature routes
O3 Source grounded adaptation	Prepared resources, dual candidates, 18/19 checks	Implemented; mathematical fidelity and live generation benchmark needed
O4 Explainable support	Attributions, guidance, CoVe and review; teacher ratings	Implemented; independent style explanation evaluation remains
O5 Adaptive assessment	Ten MCQ schema, marking and pilot use	Implemented; broader item correctness review needed
O6 Integrated validation	Component tests and limited hosted use	Implemented; complete live regression and controlled outcomes needed

The common integration objective is partly achieved. Producer and consumer paths exist, local behavior is tested, and selected hosted use is recorded. However, there is no single controlled evaluation of the complete four-component pipeline demonstrating improved attainment or reduced mental effort. The defensible outcome is an implemented research prototype with measurable component findings and preliminary human feedback. Broader effectiveness remains a future objective rather than an assumed consequence of successful integration.

6.6 Integrated limitations and threats

Construct validity is constrained by behavioral ambiguity, questionnaire-derived categories, and grouped self-report. Paas ratings provide a practical reference for effort but do not independently identify intrinsic and extraneous load. Webcam-derived gaze is approximate and sensitive to lighting, camera position, calibration, and task design. Prediction confidence is not an independently calibrated probability of psychological truth. LIME and SHAP explain model responses, not the cause of cognition.

Profiling exclusions require overlap reconciliation, and the SVM research description must be aligned with the deployed style-model family. Quantitative findings belong to the evaluated version and cannot validate an unlinked artifact.

Implementation threats include differing watch-time calculations: the frontend subtracts idle and paused duration, while raw extraction subtracts idle duration. Idle intervals crossing window boundaries also require careful treatment. Course-level identifiers used as lesson IDs can merge contexts unless documented consistently. The older explanation database path, synchronous dispatch delays, in-memory queue, and inconsistent client endpoint configuration further complicate integration.

Content validity is constrained by imperfect transcription and extraction, permissive readiness, sampling, prompt truncation, and similarity proxies that do not establish entailment. Canonical equation copying can preserve an erroneous extract. Structural MCQ validation cannot establish factual correctness or uniqueness, while verification cannot correct missing or inaccurate evidence snapshots automatically. Human oversight mitigates some errors but requires its own workload and effectiveness evaluation.

External validity is limited by two load-study videos, restricted profiling conditions, three expert evaluators, and one ten-learner classroom session. Repeated ratings and windows should be analyzed with their clustering. There are no confidence intervals or controlled learning-outcome results for the integrated platform. *Table 16* links the principal risks to practical corrective actions.

Table 16 Integrated risks and priority corrective actions

Risk	Consequence	Priority action
Model version mismatch	Study result may not describe live artifact	Archive and evaluate the deployed model
Feature definition drift	Equivalent behavior produces different inputs	Use shared fixtures and a canonical calculation
Identifier ambiguity	Histories may merge course and lesson contexts	Define student, course, lesson and subsection semantics

Risk	Consequence	Priority action
Extraction corruption	Generated content copies damaged equations	Review knowledge and validate formula fidelity
Proxy based selection	Unsupported statements may rank highly	Add claim checks and a reject or review outcome
Uneven access protection	Learner records may be exposed	Review every service route and eliminate fallback secrets
In memory work queue	Duplicate or lost work during scaling	Introduce durable ownership and coordinated retries
Small descriptive studies	Perception may be mistaken for efficacy	Controlled broader evaluation with clustered analysis

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

Lumora's value proposition is a connected route from educator resources and learner interactions to understandable, reviewable support. Teachers can reuse prepared knowledge, inspect model evidence, review learner actions, and obtain a starting point for next-lesson planning and assessment. Students can access original material, personalized explanations, progress, and approved guidance through a common interface.

Potential institutional beneficiaries include universities, schools, and training providers using video-supported learning. The undergraduate studies relate most directly to higher education, while the classroom pilot suggests a possible school use case requiring separate validation. The system may reduce repeated preparation or help focus teacher attention, but time savings, cost savings, and willingness to pay were not measured. Commercial value must therefore be investigated through real deployment and workflow evidence rather than inferred from feature count.

7.2 Market alternatives and differentiation

The relevant alternatives are conventional course delivery systems, analytics dashboards, general-purpose content generators, standalone profiling tools, and automatic question-generation services. Each can address part of the learning workflow. Lumora combines learner context, temporary load estimates, educator knowledge, local explanations, and human review within a common course experience.

Its proposed differentiation lies in traceability and integration: adaptive outputs are linked to prepared lesson material; teacher explanations retain optional technical evidence; and guidance includes review states and feedback. The two-candidate selector also records its scoring mode. These mechanisms provide an inspectable workflow, but they are not a demonstrated competitive advantage over specific commercial products. Current features, cost, privacy, accessibility, and outcome comparison is needed before market-position claims or procurement decisions.

7.3 Business model and feasibility

Potential sustainability models include institution-hosted deployment with technical support, a managed subscription, or a licensing and maintenance agreement. These are options for investigation, not established revenue arrangements. The cost base includes hosting, database operation, asset storage, transcription compute, attribution compute, external generation requests, engineering maintenance, security, and educator review.

Prepared subsection knowledge can be reused across learners, reducing repeated transcription and extraction. Cached explanations also avoid unnecessary generation when the underlying evidence and model remain current. Conversely, two-provider generation and multi-stage recommendation verification can increase cost and latency. A feasibility pilot should record resource size, preparation time, requests per learner, provider costs, failed requests, and teacher-review time. Pricing should follow those observations rather than an arbitrary estimate.

Technical feasibility is supported by implemented services and preliminary hosted use. Operational feasibility remains conditional on durable processing, consistent schemas, model provenance, and support staffing. Adoption also requires an institution to define who approves generated material, how students can challenge an interpretation, and what happens when an external provider changes its service. These responsibilities are part of the operating model rather than incidental configuration.

7.4 Adoption intellectual property and compliance

Adoption requires clear resource ownership and permission to upload, transform, and store teaching content. Team-created code, model artifacts, research datasets, third-party libraries, provider terms, and branding may carry different rights. An institution should establish the applicable ownership and licensing arrangements before distributing or commercializing the system. The report does not assert legal clearance for every uploaded resource or dependency.

Privacy arrangements should specify behavioral data purpose, access, retention, deletion, consent, and external disclosure. Prompted lesson text and learner-state labels may be transmitted to model providers. It-related processing needs an explicit explanation of what is processed and retained; derived-feature collection does not by

itself establish that all records are anonymous. School deployment additionally requires institutional safeguarding arrangements and an appropriate consent process for minors. Current legal and policy compliance requires a deployment-specific review rather than a blanket claim.

Responsible adoption preserves teacher and learner agency. Guidance should be amendable, rejected advice should be withheld, and technical evidence should remain available for review. Students should retain access to varied content rather than being confined by a predicted category. Model outputs should not determine grades, diagnose learners, or be treated as permanent ability labels. UNESCO's human-centered principles support these design priorities [19].

7.5 Sustainable Development Goal contribution

The plausible contribution to SDG 4 is improved access to responsive learning support and better use of educational resources [2]. Browser interaction data avoids specialized load-sensing devices, and prepared knowledge can be reused across classes. Teacher review creates a pathway for applying analytical results within instructional practice rather than leaving them as isolated dashboard indicators.

Possible negative effects include bandwidth and webcam barriers, additional teacher workload, misleading classifications, and energy and financial costs from repeated AI requests. These should be measured alongside educational benefits. Accessible alternatives, local caching, economical generation policies, and teacher training can support sustainability. A future impact evaluation should examine participation across device and language groups, learning outcomes, review burden, and cost per completed lesson. No present result establishes a population-level SDG impact.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The project addresses the gap between digital learning activity and actionable educational support through an integrated platform. Learner profiling supplies contextual information, two-minute behavioral windows supply changing load estimates, resource preparation supplies instructional evidence, and explainable services produce reviewable guidance, planning, and assessment. The common result is a connected research prototype whose outputs can be traced to learner records, resource context, and service decisions.

The strongest quantitative model finding is the load classifier's 79.52% accuracy and 79.55% macro F1 on 293 held-out windows from a participant-separated study. Technical verification supports selected component behavior, while the mathematical fixture failure identifies a substantive content-integrity issue. Three teachers gave favorable preliminary comparative ratings to guidance and adaptive MCQs, and ten classroom learners completed a hosted lesson and quiz. These results support feasibility and limit perceived usefulness.

They do not establish the accuracy of every deployed model, universal extraction fidelity, or improved learning from the complete pipeline. Differences between the profiling experiment and operational artifact, feature calculation routes, and separate evaluation populations remain important.

8.2 Principal integrated contribution

The principal contribution is the traceable relationship between learner evidence, source knowledge, and educational response. Each stage has a defined output: a profile, an accepted or skipped prediction, a prepared knowledge representation, an attribution record, a generated support draft, or a review decision. Keeping these outputs distinct allows an error to be examined at its origin rather than treating an attractive final explanation as proof that the entire process is correct.

A second contribution is the task-specific handling of explanation and review. Style explanations preserve influential signed evidence without advice, load guidance uses a positive-support list for a defined purpose, and class recommendations verify claims

against structured evidence. Quiz validation remains an independent structural check. Together, these choices show how established models and AI techniques can be organized within an educator-controlled workflow. The novelty lies in the implementation and its investigated integration boundaries, rather than in inventing Random Forest, SVM, LIME, SHAP, or generative language modelling.

8.3 Future directions

The priority is evidence consistency. Archive final datasets, exclusion logs, participant splits, fitted preprocessing, and model signatures. Reconcile the SVM study with the operational style artifact and evaluate the deployed version independently. Align frontend and server feature calculations, document course and lesson identifier semantics, and test boundary-crossing idle periods. Add confidence calibration and participant-aware uncertainty analysis for load predictions.

The second priority is content integrity. Introduce review of extracted knowledge and equations before treating a lesson as verified. Retain quality warnings rather than overriding them with non-empty-content readiness. Benchmark diverse PDFs, presentations, scanned material, diagrams, accents, and technical language. Evaluate claim support, concept coverage, omissions, numeric values, and equation identity in actual generated outputs. Compare selection modes and weights against independent teacher judgment, including a reject-or-review option.

The third priority is operational reliability. Replace or coordinate in-memory processing with durable job ownership, retries, timeouts, and bounded concurrency. Reconcile analytical schemas, add migrations and cross-service contract tests, standardize endpoint configuration, and strengthen authorization. Preserve generation versions, prompts, review history, and trace identifiers. Test concurrent requests, provider outages, backups, restoration, and accessible alternatives to webcam tasks.

Finally, conduct a controlled, longer-term educational study after the technical issues are addressed. Compare the integrated workflow with a defined baseline using prior-knowledge measures, pre/post assessment, delayed retention, self-reported effort, and teacher workload. Recruit broader subjects and institutions and account for repeated observations. Evaluate cognitive-style explanations separately from load guidance and

audit generated answer keys item by item. These steps would establish whether the connected workflow produces educational benefit beyond its present technical feasibility.

REFERENCES

[1] J. Sweller, “Cognitive load during problem solving: Effects on learning,” Cognitive Science, vol. 12, no. 2, pp. 257–285, 1988, doi: 10.1207/s15516709cog1202_4.

[2] United Nations Department of Economic and Social Affairs, “Goal 4,” Sustainable Development Goals. Accessed: Oct. 11, 2026. [Online]. Available: https://sdgs.un.org/goals/goal4

[3] N. Entwistle, H. Tait, and V. McCune, “Patterns of response to an approaches to studying inventory across contrasting groups and contexts,” European Journal of Psychology of Education, vol. 15, no. 1, pp. 33–48, 2000, doi: 10.1007/BF03173165.

[4] I. Choi, M. Koo, and J. A. Choi, “Individual differences in analytic versus holistic thinking,” Personality and Social Psychology Bulletin, vol. 33, no. 5, pp. 691–705, 2007, doi: 10.1177/0146167206298568.

[5] M. Merkt, A. Hoppe, G. Bruns, R. Ewerth, and M. Huff, “Pushing the button: Why do learners pause online videos?,” Computers & Education, vol. 176, Art. no. 104355, 2022, doi: 10.1016/j.compedu.2021.104355.

[6] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, “On faithfulness and factuality in abstractive summarization,” in Proc. 58th Annu. Meeting Assoc. Computational Linguistics, 2020, pp. 1906–1919, doi: 10.18653/v1/2020.acl-main.173.

[7] H. Pashler, M. McDaniel, D. Rohrer, and R. Bjork, “Learning styles: Concepts and evidence,” Psychological Science in the Public Interest, vol. 9, no. 3, pp. 105–119, 2008, doi: 10.1111/j.1539-6053.2009.01038.x.

[8] C. Cortes and V. Vapnik, “Support-vector networks,” Machine Learning, vol. 20, pp. 273–297, 1995, doi: 10.1007/BF00994018.

[9] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” Journal of Artificial Intelligence Research, vol. 16, pp. 321–357, 2002, doi: 10.1613/jair.953.

[10] F. G. W. C. Paas, “Training strategies for attaining transfer of problem-solving skill in statistics: A cognitive-load approach,” Journal of Educational Psychology, vol. 84, no. 4, pp. 429–434, 1992, doi: 10.1037/0022-0663.84.4.429.

[11] L. Breiman, “Random forests,” Machine Learning, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[12] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in Proc. EMNLP-IJCNLP, 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410.

[13] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, “MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers,” in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 5776–5788.

[14] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why should I trust you?: Explaining the predictions of any classifier,” in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 1135–1144, doi: 10.1145/2939672.2939778.

[15] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Advances in Neural Information Processing Systems, vol. 30, 2017, pp. 4765–4774.

[16] J. Hattie and H. Timperley, “The power of feedback,” Review of Educational Research, vol. 77, no. 1, pp. 81–112, 2007, doi: 10.3102/003465430298487.

[17] J. D. Karpicke and H. L. Roediger III, “The critical importance of retrieval for learning,” Science, vol. 319, no. 5865, pp. 966–968, 2008, doi: 10.1126/science.1152408.

[18] S. Dhuliawala et al., “Chain-of-Verification reduces hallucination in large language models,” in Findings of the Association for Computational Linguistics: ACL 2024, 2024, pp. 3563–3578, doi: 10.18653/v1/2024.findings-acl.212.

[19] F. Miao and W. Holmes, Guidance for Generative AI in Education and Research. Paris, France: UNESCO, 2023, doi: 10.54675/EWZM9535.

APPENDICES

Appendix A Integrated interface specification

The interface summary in *Table 17* identifies the purpose of each main gateway route and the record it produces or consumes. Paths are grouped by workflow rather than assumed to share one permission policy. Producer and consumer services should agree on identifier meaning, required fields, error semantics, and model version before a release.

Table 17 Gateway interface responsibilities

Route group	Producer or consumer	Principal output
/api/auth and /api/enrollments	Core backend	Identity, approval and enrolment
/cognitive-style	Profile backend	Questionnaire and interaction profile records
/api/cognitive-load/predict	Load API	Reliable or skipped prediction
/api/cognitive-load/events/raw	Load API	Saved event and automation status
/api/cognitive-load/.../load-trend	Load API	Current state, timeline and trend
/api/courses and /api/sections	Resource service	Course structure and preparation status
/api/gpt and /api/deepseek	Candidate services	Prompt, content candidates and selection
/api/lime-ai/v1	LIME and guidance	Predictions, summaries and reviewed guidance
/api/shap-ai/v1	SHAP service	Stored record attribution
/api/cognitive-style-ai/v1	Style explanation	Student level cached analysis
/api/next-lesson-recommendation	Recommendation service	Course evidence, plan and review

Route group	Producer or consumer	Principal output
/api/exam	Backend to exam service	Materials, generated quiz and marking

The load feature contract contains six ordered numeric fields. Four represent counts: pauses, forward navigation, backward seeks, and rate changes. Two represent seconds: idle duration and time on content. Paused duration is a reliability field rather than a classifier input. Student, lesson, session, and window identifiers are linkage fields rather than predictors. The target score maps 1 through 5 to Very Low through Very High. *Table 18* records the operational definitions and consistency issues.

Table 18 Six feature inference contract and reliability field

Field	Unit and definition	Consistency requirement
pause_frequency	Count of confirmed video pauses	Exclude seeking related temporary pauses
navigation_count_video	Count of forward seeks	Use the same direction convention
rewatch_segments	Count of backward seeks	Do not interpret as verified comprehension difficulty
playback_rate_change	Count of speed adjustments	Match collection and inference definitions
idle_duration_video	Accumulated inactive seconds	Handle intervals across window boundaries
time_on_content	Effective content seconds	Align frontend and raw extraction formulas
paused_duration_video	Paused seconds; gate only	Keep separate from the six model inputs

For a reliable, saved prediction, the expected sequence is validation, feature ordering, inference, feature-window storage, prediction-log storage, CSV recording, and dispatch. A skipped prediction should not acquire a valid class. Style analysis should

reuse a completed profile only when its model signature remains current. Guidance sharing requires an explicit review transition. MCQ responses should omit correct answers during an attempt and return marking information after submission. These invariants are suitable targets for a shared integration regression.

Appendix B Team contribution declaration

The responsibility allocation follows the four-component system flow documented in Table 4. The members should confirm the final common report, their responsibility boundaries, and the shared integration work before signing. The following record is intentionally unsigned; it does not imply that confirmation or supervisor certification has already occurred.

AI assistance was used to consolidate technical descriptions, organize the common report, improve academic language, check bibliographic records, and prepare the editable document. Research measurements and evaluation outcomes remain distinct from this drafting assistance. Responsibility for submission, contribution confirmation, institutional disclosure requirements, and the accuracy of the final report remain with the research team.

Appendix C Extended evaluation records

Table 19 preserves the anonymized classroom observations used to calculate the pilot summary. The 78.0% aggregate score follows from 78 correct responses divided by 100 attempted responses. The 4.30/5 clarity mean follows from a total rating of 43 across ten learners. These are descriptive results from one post-lesson assessment, not estimates of improvement.

Table 19 Anonymized classroom learner observations

Learner	Load	Correct /10	Score (%)	Clarity /5
S01	Low	8	80	3
S02	High	7	70	4
S03	High	9	90	5
S04	Medium	9	90	5

Learner	Load	Correct /10	Score (%)	Clarity /5
S05	Very High	8	80	4
S06	High	7	70	4
S07	High	8	80	5
S08	High	7	70	4
S09	Medium	7	70	4
S10	High	8	80	5
Total / mean	Mixed	78 /100	78.0	4.30

Table 20 separates the interpretation of the numerical outcomes. Automated passes, classification metrics, teacher ratings, and classroom scores have different denominators and must retain those denominators in subsequent publications. The teacher-guidance mean is based on five criteria across nine paired cases, with repeated ratings from three teachers. The MCQ answer-key counts represent thirty item judgments per method, not thirty teacher participants.

Table 20 Evaluation denominators and interpretation

Outcome	Denominator	What it establishes
79.52% load accuracy	233 of 293 test windows	Classification agreement in studied context
79.55% macro F1	Five class F1 values	Balanced class level summary
18/19 resource checks	19 focused assertions	Selected helper behavior; one failure retained
4.84 vs 3.87 guidance	45 ratings per method across 9 paired cases	Descriptive ratings from 3 teachers
9/9 guidance preference	9 repeated scenario comparisons	Preference within evaluated scenarios

Outcome	Denominator	What it establishes
29/30 adaptive key judgments	30 judgments from 3 teachers	One adaptive item needs review
78.0% classroom score	100 answers from 10 learners	Post lesson attempt performance
4.30/5 classroom clarity	10 learner ratings	Perceived MCQ clarity in one session

The evaluation archive should preserve the instruments, counterbalanced method mapping, source material, model versions, and raw anonymized observations for future analysis. A complete end-to-end release evaluation should also retain the feature snapshot, generated candidates, selected output, attribution records, teacher decision, and quiz key for each case. This would allow disagreements and failures to be investigated without changing the population or measurements being reported.